

Q1. Smart Document Editor using String, StringBuilder and StringBuffer

CODE

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String paragraph = sc.nextLine();

        int characters = paragraph.length();

        String temp = paragraph.trim();
        int words = 0;
        if (!temp.isEmpty()) {
            words = temp.split("\\s+").length;
        }

        int sentences = 0;
        for (int i = 0; i < paragraph.length(); i++) {
            char ch = paragraph.charAt(i);
            if (ch == '.' || ch == '!' || ch == '?') {
                sentences++;
            }
        }

        String oldWord = sc.nextLine();
        String newWord = sc.nextLine();

        String edited = paragraph.replace(oldWord, newWord);

        String[] sentenceArray = edited.split("(?<=[.!?])");

        StringBuffer history = new StringBuffer();
        history.append("Replaced \""
            .append(oldWord)
            .append("\" with \""
            .append(newWord)
            .append("\"");

        StringBuilder finalDoc = new StringBuilder();
        for (int i = 0; i < sentenceArray.length; i++) {
            finalDoc.append(sentenceArray[i].trim());
            if (i != sentenceArray.length - 1) {
                finalDoc.append("\n");
            }
        }

        System.out.println("Original Document : ");
        System.out.println(paragraph);

        System.out.println("Characters : " + characters);
        System.out.println("Words : " + words);
        System.out.println("Sentences : " + sentences);

        System.out.println("Editing History : ");
        System.out.println(history);

        System.out.println("Final Document : ");
        System.out.println(finalDoc.toString());

        sc.close();
    }
}
```

```
    }  
}
```

INPUT

Java is easy. java is powerful. Java is popular.
Java
Python

OUTPUT

(no output)

Q2. Hospital Token Management System using List Interface

CODE

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        ArrayList<Integer> id = new ArrayList<Integer>();
        ArrayList<String> name = new ArrayList<String>();
        ArrayList<String> dept = new ArrayList<String>();

        int n = sc.nextInt();
        sc.nextLine();

        for (int i = 0; i < n; i++) {
            id.add(sc.nextInt());
            sc.nextLine();
            name.add(sc.nextLine());
            dept.add(sc.nextLine());
        }

        int updateId = sc.nextInt();
        sc.nextLine();
        String newName = sc.nextLine();
        String newDept = sc.nextLine();

        for (int i = 0; i < id.size(); i++) {
            if (id.get(i) == updateId) {
                name.set(i, newName);
                dept.set(i, newDept);
                break;
            }
        }

        int removeId = sc.nextInt();

        for (int i = 0; i < id.size(); i++) {
            if (id.get(i) == removeId) {
                id.remove(i);
                name.remove(i);
                dept.remove(i);
                break;
            }
        }

        sc.nextLine();
        String searchDept = sc.nextLine();

        System.out.println("Patients:");
        for (int i = 0; i < id.size(); i++) {
            System.out.println(id.get(i) + " " + name.get(i) + " " + dept.get(i));
        }

        System.out.println("Department Patients:");
        for (int i = 0; i < id.size(); i++) {
            if (dept.get(i).equalsIgnoreCase(searchDept)) {
                System.out.println(id.get(i) + " " + name.get(i));
            }
        }

        if (id.size() > 0) {
            System.out.println("First Patient:");
        }
    }
}
```



```
        System.out.println(id.get(0) + " " + name.get(0));

        System.out.println("Last Patient:");
        System.out.println(id.get(id.size() - 1) + " " + name.get(name.size() - 1));
    }

    sc.close();
}
```

INPUT

```
3
101
Rahul
Cardiology
102
Priya
Neurology
103
Kiran
Cardiology
102
Priya Sharma
Neurology
103
Cardiology
```

OUTPUT

(no output)

Q3. Airport Boarding Queue Management using Queue Interface

CODE

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        Queue<String> queue = new LinkedList<String>();
        PriorityQueue<String> vip = new PriorityQueue<String>();

        int n = sc.nextInt();
        sc.nextLine();

        for (int i = 0; i < n; i++) {
            queue.offer(sc.nextLine());
        }

        int m = sc.nextInt();
        sc.nextLine();

        for (int i = 0; i < m; i++) {
            vip.offer(sc.nextLine());
        }

        System.out.println("Current Queue:");
        for (String s : queue) {
            System.out.println(s);
        }

        System.out.println("VIP Queue:");
        for (String s : vip) {
            System.out.println(s);
        }

        if (!queue.isEmpty()) {
            System.out.println("Next Passenger:");
            System.out.println(queue.peek());
        }

        while (!vip.isEmpty()) {
            System.out.println("Boarded VIP: " + vip.poll());
        }

        while (!queue.isEmpty()) {
            System.out.println("Boarded: " + queue.poll());
            System.out.println("Remaining Queue:");
            if (queue.isEmpty()) {
                System.out.println("Empty");
            } else {
                for (String s : queue) {
                    System.out.println(s);
                }
            }
        }

        sc.close();
    }
}
```

INPUT

```
3
Rahul
Priya
Kiran
```

2
Anitha
Vijay

OUTPUT (no output)

Q4. Digital Certificate Verification using Set Interface

CODE

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        LinkedHashSet<String> certificates = new LinkedHashSet<String>();

        int n = sc.nextInt();
        sc.nextLine();

        for (int i = 0; i < n; i++) {
            certificates.add(sc.nextLine());
        }

        String search = sc.nextLine();

        if (certificates.contains(search))
            System.out.println("Certificate Found");
        else
            System.out.println("Certificate Not Found");

        System.out.println("Certificates in Insertion Order:");
        for (String s : certificates)
            System.out.println(s);

        System.out.println("Certificates in Sorted Order:");
        TreeSet<String> sorted = new TreeSet<String>(certificates);
        for (String s : sorted)
            System.out.println(s);

        System.out.println("Total Unique Certificates: " + certificates.size());

        String revoke = sc.nextLine();
        certificates.remove(revoke);

        System.out.println("Certificates After Revocation:");
        for (String s : certificates)
            System.out.println(s);

        sc.close();
    }
}
```

INPUT
5
C101
C102
C103
C102
C104
C103
C102

OUTPUT (no output)

Q5. Smart Inventory Management using Map Interface

CODE

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        HashMap<Integer, String> name = new HashMap<Integer, String>();
        HashMap<Integer, Integer> quantity = new HashMap<Integer, Integer>();
        HashMap<Integer, Double> price = new HashMap<Integer, Double>();
        HashMap<Integer, String> category = new HashMap<Integer, String>();

        int n = sc.nextInt();

        for (int i = 0; i < n; i++) {
            int id = sc.nextInt();
            String pname = sc.next();
            int qty = sc.nextInt();
            double pr = sc.nextDouble();
            String cat = sc.next();

            name.put(id, pname);
            quantity.put(id, qty);
            price.put(id, pr);
            category.put(id, cat);
        }

        int updateId = sc.nextInt();
        int newQty = sc.nextInt();

        if (quantity.containsKey(updateId)) {
            quantity.put(updateId, newQty);
        }

        int searchId = sc.nextInt();

        if (name.containsKey(searchId)) {
            System.out.println("Product Found");
            System.out.println(searchId + " " + name.get(searchId) + " " + quantity.get(searchId) + " " + price.get(searchId) + " " + category.get(searchId));
        } else {
            System.out.println("Product Not Found");
        }

        double total = 0;

        for (Integer id : name.keySet()) {
            total += quantity.get(id) * price.get(id);
        }

        System.out.println("Total Inventory Value: " + total);

        String searchCategory = sc.next();

        System.out.println("Category Products:");
        for (Integer id : name.keySet()) {
            if (category.get(id).equalsIgnoreCase(searchCategory)) {
                System.out.println(id + " " + name.get(id));
            }
        }

        int removeId = sc.nextInt();
    }
}
```



```
name.remove(removeId);
quantity.remove(removeId);
price.remove(removeId);
category.remove(removeId);

System.out.println("Available Products:");
for (Integer id : name.keySet()) {
    System.out.println(id + " " + name.get(id) + " " + quantity.get(id) + " " + price.get(id) + " " + category.get(id));
}

sc.close();
}
```

INPUT

```
3
101 Laptop 5 50000 Electronics
102 Mouse 10 500 Electronics
103 Chair 4 2500 Furniture
102 15
101
Electronics
103
```

OUTPUT

(no output)